New excluded minors for the class $\mathcal{M}_3$ of regular matroids

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Abstract

Engel, de Gaay Fortman, and Schreieder attach to each prime ℓ a minor-closed class $\mathcal{M}_\ell$ of regular matroids, whose excluded minors govern the failure of the integral Hodge conjecture for curve classes on very general principally polarized abelian varieties. For $\ell = 2$ the class is exactly the cographic matroids, with excluded minors $M(K_5)$ and $M(K_{3,3})$ by Tutte's theorem; for the odd prime $\ell = 3$ the only excluded minor known so far is $M(K_{3,5})$ [EGFS, Prop. 8.11], and the general characterization is [EGFS, Problem 8.8]. We exhibit five new excluded minors for $\mathcal{M}_3$, of ranks 8, 9, 9, 9, and 10. None contains $M(K_{3,5})$ or any of the other four as a minor, so the excluded-minor list of Problem 8.8 has at least six members, with excluded minors at every rank from 7 through 10. The list is structurally diverse: the rank-8 and rank-10 minors are graphic matroids of bipartite apex constructions with automorphism group S_4 (the 3-cube plus a one-sided apex, and the subdivided K_4 plus an apex), whereas the three rank-9 minors are graphic matroids of non-bipartite, non-planar graphs with automorphism groups of order at most 4 and no such apex structure. Every verdict is certified by a finite $\mathbb{F}_3$-linear-algebra computation with explicit machine-checkable witnesses, cross-checked by independent implementations. Combined with a recent result of Engel and Schreieder [ES, Cor. 2.10], the certificates further show that the five matroids lie outside the larger class $\widetilde{\mathcal{M}}_3$ of [EGFS, Rmk. 8.5]. We further record the first $\ell = 5$ data on this corank-8 slice: $M(K_{3,5})$ and the rank-8 and rank-10 minors lie in $\mathcal{M}_5$, each by an explicit certificate, so none of these three contributes a factor of 5 to its radical distance; the $\ell = 5$ status of the three rank-9 minors has not yet been computed. This complements the recent theorem of Engel and Schreieder [ES] that the minimal algebraic multiple of the minimal class on a very general principally polarized abelian variety is divisible by all primes $p \leq (k+1)/2$, which resolves [EGFS, Problem 8.13].

1 Setup

We work in the framework of [EGFS]. To a regular matroid (R, S) of rank g on ground set S, with a totally unimodular integral realization $S \hookrightarrow U^*$ (equivalently a matrix $A = (I_g \mid D) \in \mathbb{Z}^{g\times|S|}$, whose row space $U = U(R) \subset \mathbb{Z}^S$ is the cut lattice), [EGFS] associate the following combinatorial data.

The Albanese graph [EGFS, Def. 5.3]. For non-negative integers $j \leq r$, the (ℓ^r, ℓ^j)-Albanese graph $\mathrm{Alb}_{\ell^r,\ell^j}(R)$ is the oriented S-colored graph with vertex set

$$V \;=\; \mathbb{Z}^S/(\ell^j U + \ell^r \mathbb{Z}^S),$$

in which $[v]$ and $[w]$ are joined by an oriented edge of color $s \in S$, directed from $[v]$ to $[w]$, precisely when $[w] = [v+e_s]$. We write $\mathrm{Alb}_\ell(R) := \mathrm{Alb}_{\ell^1,\ell^0}(R)$, whose vertex set is $\mathbb{F}_\ell^S/\bar{U}$ (with $\bar{U} = U \bmod \ell$, of dimension g), so that $|V| = \ell^{|S|-g}$ and each vertex has one out-edge of each color.

The color profile [EGFS, Def. 5.1] is the linear map $\lambda\colon C_1(\mathrm{Alb}_\ell(R),\Lambda)\to\Lambda^S$ sending an oriented edge of color s to the basis vector e_s; for a 1-chain b we write $\lambda_s(b)$ for its s-component.

Solutions and membership [EGFS, Def. 5.8, Def. 8.3]. A Λ-solution of (R,S) in a graph G is a family $(b_s)_{s\in S}$ of 1-chains, b_s supported on the color-s edges, such that $\sum_s c_s\, b_s$ is closed whenever $\sum_s c_s e_s \in U_\Lambda$. The matroid R lies in the class

$$\mathcal{M}_\ell \;:=\; \big\{\, R \text{ regular } : \; R \text{ admits a } \mathbb{Z}/\ell\text{-solution in } \mathrm{Alb}_\ell(R) \text{ with constant nonzero color profile} \,\big\},$$

i.e. a solution with $\lambda(b_s)=c$ for a single $c\in(\mathbb{Z}/\ell)^\times$ and all s.

By [EGFS, Prop. 7.2] each $\mathcal{M}_\ell$ is closed under taking minors; within graphic matroids, the Robertson–Seymour graph-minor theorem therefore guarantees a characterization by finitely many excluded graph minors. For the full class of regular matroids finiteness is expected but open — $\mathcal{M}_\ell$ "is expected to be determined by a finite list of excluded minors" via the conjectural extension of the Robertson–Seymour theorem to regular matroids [EGFS, §8.5] — and Problem 8.8 below asks for the characterization. Two facts frame the present note.

- $\ell = 2$ **is classical.** $\mathcal{M}_2$ is the class of cographic matroids [EGFS, Thm. 7.1]; a graphic matroid $M(G)$ is cographic iff G is planar. Tutte's theorem gives the two excluded minors $M(K_5)$ and $M(K_{3,3})$.
- $\ell = 3$ **had, until now, a single known excluded minor.** $M(K_{3,5})$ (rank 7) satisfies $M(K_{3,5})\notin\mathcal{M}_3$ while every single-element minor lies in $\mathcal{M}_3$, hence is an excluded minor for $\mathcal{M}_3$ [EGFS, Prop. 8.11]. As of both [EGFS] and the subsequent work [ES] of Engel and Schreieder, $M(K_{3,5})$ remains the only excluded minor explicitly identified for $\mathcal{M}_3$ in the literature: [ES] proves the further non-memberships $M(K_{2p+1})\notin\mathcal{M}_p$ and $M(K_{3,2p-1})\notin\mathcal{M}_p$ for every prime p (its Theorems 1.5 and 1.7; the case $p=3$ recovers $M(K_7)\notin\mathcal{M}_3$ — first established in the follow-up [EGFS2, Thm. 1.2] — and $M(K_{3,5})\notin\mathcal{M}_3$), but it identifies no excluded minors and leaves Problem 8.8 open. The corresponding open problem is:

> **Problem 8.8 [EGFS].** *Let ℓ be an odd prime. Characterize the subclass $\mathcal{M}_\ell$ of the class of regular matroids via a finite list of excluded minors.*

The radical distance $d(R) := \mathrm{lcm}\{\ell \text{ prime} : R\notin\mathcal{M}_\ell\}$ [EGFS, Def. 8.9] measures how far R is from cographic; it is finite and divides the product of the primes below $\mathrm{rank}(R)$, and $d(R)=1$ iff R is cographic. Excluded minors for $\mathcal{M}_\ell$ have rank $>\ell$ [EGFS, Rmk. 8.6 / Thm. 1.10].

2 The results

We describe five graphs $G_1,\dots,G_5$; all are non-planar of girth 4 with vertex connectivity exactly 3, and each graphic matroid $M(G_i)$ is regular, simple, and of corank 8.

The rank-8 graph G_1. Let G_1 be the bipartite graph on parts $X=\{1,2,3,4\}$ and $Y=\{5,6,7,8,9\}$ with edge set

$$E(G_1) = \{\, 15, 25, 35, \quad 16, 26, 46, \quad 17, 37, 47, \quad 28, 38, 48, \quad 19, 29, 39, 49 \,\}.$$

Equivalently, $G_1 = K_{4,5}$ with the perfect matching $\{18, 27, 36, 45\}$ deleted (each vertex of X loses one edge, to four of the five vertices of Y; vertex 9 remains complete to X). It has 9 vertices, 16 edges, degree sequence $(4^5, 3^4)$, girth 4, is 3-connected and non-planar, with automorphism group of order 24, isomorphic to S_4. Its graph6 code is `H?zTbbo` (canonical form `H?Bmtpw`). There is a second useful description: deleting the unique degree-4 vertex of Y, namely 9, leaves the 3-regular bipartite graph on $\{1,2,3,4\} \cup \{5,6,7,8\}$, which is $K_{4,4}$ minus a perfect matching, i.e. the 3-cube Q_3. Thus

$$G_1 \;=\; Q_3 \text{ together with an apex joined to all four vertices of one side of } Q_3.$$

$M(G_1)$ has rank 8 and size 16.

The rank-10 graph G_2. Let G_2 be the graph obtained from K_4 by subdividing each of its six edges once and joining a new apex vertex to the six subdivision vertices: 11 vertices, 18 edges, bipartite with parts of sizes 6 and 5, degree sequence $(6^1, 3^{10})$, girth 4, non-planar, automorphism group S_4 (order 24). Its graph6 code is `J??FCpSJFw?` (canonical form `J?hcab?K?^_`). $M(G_2)$ has rank 10 and size 18.

The three rank-9 graphs G_3, G_4, G_5. Each is a graph on 10 vertices with 17 edges, degree sequence $(4^4, 3^6)$, girth 4, *non-bipartite* and non-planar; each graphic matroid has rank 9 and size 17. On vertex set $\{0,\dots,9\}$:

$$\begin{aligned}
E(G_3) &= \{04, 06, 07,\; 15, 16, 17,\; 26, 27, 28,\; 37, 38, 39,\; 48, 49,\; 58, 59,\; 69\},\\
E(G_4) &= \{05, 07, 08,\; 15, 17, 18,\; 26, 27, 29,\; 36, 38, 39,\; 47, 48, 49,\; 56, 59\},\\
E(G_5) &= \{03, 05, 06,\; 14, 16, 18,\; 25, 26, 27,\; 37, 38, 39,\; 47, 49,\; 58, 59,\; 69\}.
\end{aligned}$$

Their graph6 codes are ``I?`FF_{F_`` (canonical form, computed with `bliss`, `I@?LQngt?`), `I?B@nRWN?` (canonical `I??[rNoy?`), and `ICQf@pSF_` (canonical `IAGWMUqw_`); the three graphs are pairwise non-isomorphic. Their automorphism groups are small: $C_2 \times C_2$, $C_2 \times C_2$, and C_2 respectively. Unlike G_1 and G_2, the three rank-9 graphs do not present themselves as a small structured base plus an apex vertex; they were found by unstructured census search rather than by construction.

Theorem 2.1. *Each of the five graphic matroids $M(G_1), \dots, M(G_5)$ is an excluded minor for the class $\mathcal{M}_3$, and no one of the six matroids $M(K_{3,5}), M(G_1), \dots, M(G_5)$ contains another as a minor. Moreover each $M(G_i)$ lies outside the larger class $\widetilde{\mathcal{M}}_3$ of [EGFS, Rmk. 8.5], and is an excluded minor for $\widetilde{\mathcal{M}}_3$ as well.*

The non-memberships, the minimality, and the pairwise independence are established by explicit certificates in §3, where the $\widetilde{\mathcal{M}}_3$-strengthening is also proved.

Corollary 2.2. *The class $\mathcal{M}_3$ is not characterized among regular matroids by the exclusion of $M(K_{3,5})$ alone. Any finite excluded-minor list for $\mathcal{M}_3$ (Problem 8.8) contains at least six members: $M(K_{3,5})$ and $M(G_1), \dots, M(G_5)$.*

Corollary 2.3. *Excluded minors for $\mathcal{M}_3$ occur at every rank from 7 to 10: $M(K_{3,5})$ has rank 7, and the five matroids above have ranks $8, 9, 9, 9, 10$.*

Structural diversity. The known excluded-minor list is not the work of a single construction. G_1 and G_2 are bipartite base-plus-apex graphs with automorphism group S_4; the three rank-9 graphs are non-bipartite, admit no such decomposition, and have automorphism groups of order at most 4. In the systematic search that produced these results (§5), 214 constructed base-plus-apex candidates over 24 distinct bases yielded *zero* non-members of $\mathcal{M}_3$, while the undirected census of generic corank-8 graphs produced the three rank-9 excluded minors. On the present data, a description of the excluded-minor list of Problem 8.8 by a single structural family is not supported.

Consequences for radical distance. These witnesses give no new lower bound on the radical-distance function. Each G_i is non-planar, so each $M(G_i)$ is non-cographic and $M(G_i) \notin \mathcal{M}_2$; together with $M(G_i) \notin \mathcal{M}_3$ this yields $6 \mid d\big(M(G_i)\big)$, and hence $6 \mid d(g)$ for $g = 8, 9, 10$, where $d(g) := \operatorname{lcm}\{d(R) : R \text{ regular of rank } g\}$. But $6 \mid d(g)$ is already known for all $g \geq 7$ from $M(K_{3,5}) \preceq M(K_{3,n})$ (rank $n+2$). Thus the five matroids are independent witnesses for known divisibilities, and contribute no new numerical floor. A genuinely new floor at this corank would require one of them to lie outside $\mathcal{M}_5$. For $M(G_1)$ and $M(G_2)$ this has since been tested and fails: both lie in $\mathcal{M}_5$ (§4, by explicit certificates). For the three rank-9 matroids the $\ell = 5$ computation has not yet been carried out, and we make no claim about it.

3 The certificates and the method

A linear-algebra formulation of membership. Scaling the constant profile to $c = 1$, [EGFS, Def. 8.3] reads: $R \in \mathcal{M}_\ell$ iff the single inhomogeneous $\mathbb{F}_\ell$-linear system

$$\text{(closedness)} \quad \textstyle\sum_s u_s \, \partial b_s = 0 \;\text{ for a basis } \{u\} \text{ of } \bar{U}, \qquad \text{(profile)} \quad \lambda_s(b) = 1 \;\text{ for all } s \in S,$$

in the unknown 1-chains $(b_s)_{s \in S}$ is solvable. Solvability is decided by rank comparison during the search; the certificates themselves use no rank computation. Membership is certified by an explicit solution (a *primal witness*), and non-membership by an explicit *dual certificate* — a left-kernel witness y with $y^\mathsf{T} M = 0$ and $y^\mathsf{T} b \neq 0$ (a Farkas-type alternative over $\mathbb{F}_3$) — each checkable by direct matrix–vector arithmetic over $\mathbb{F}_3$.

Non-membership. Write C for the closedness matrix ($g \cdot 3^8$ rows, one per basis element of $\bar{U}$ and Albanese vertex) and L for the color-profile map (n rows), both acting on the space of 1-chain families ($n \cdot 3^8$ coordinates, one per colored Albanese edge, with $|V| = 3^8 = 6561$); membership is the solvability of

$$Mx = b, \qquad M = \begin{pmatrix} C \\ L \end{pmatrix}, \qquad b = (0, \ldots, 0 \mid 1, \ldots, 1)^\mathsf{T}.$$

For each of the five matroids, a verified dual certificate establishes $M(G_i) \notin \mathcal{M}_3$ on this full membership system in $\mathrm{Alb}_3(M(G_i))$; the recorded ranks show the infeasibility signature $\operatorname{rank}[M \mid b] = \operatorname{rank} M + 1$ in every case:

matroid	(g, n)	rows of M	columns of M	rank M	rank $[M \mid b]$
$M(G_1)$	$(8, 16)$	52,504	104,976	48,274	48,275
$M(G_2)$	$(10, 18)$	65,628	118,098	58,230	58,231
$M(G_3)$	$(9, 17)$	59,066	111,537	53,591	53,592
$M(G_4)$	$(9, 17)$	59,066	111,537	53,555	53,556
$M(G_5)$	$(9, 17)$	59,066	111,537	53,612	53,613

For $M(G_1)$ and $M(G_2)$ the recorded ranks exhibit the failure in its strongest form: $\operatorname{rank} C = \operatorname{rank} M$ (48,274 and 58,230 respectively, with closedness solution spaces of dimensions 56,702 and 59,868). That is, the profile rows of L are linear combinations of rows of C, so *every* $\mathbb{Z}/3$-solution of the closedness equations has identically zero color profile; in particular no solution with constant profile 1 exists. For $M(G_3), M(G_4), M(G_5)$ the rank of C alone was not separately recorded, and we state only the tabulated pair. In all five cases the dual certificate, not the rank computation, carries the claim.

Minimality and pairwise independence (one argument). For each of the five matroids, *every* single-element minor (each deletion and each contraction) was verified to lie in $\mathcal{M}_3$ by an explicit primal witness. For $M(G_1)$ and $M(G_2)$, whose automorphism groups act with four orbits on the 32 (resp. 36) single-element minors, one representative per orbit was certified (for $M(G_1)$: the four orbit classes of sizes $12+12+4+4$, deletions and contractions of cube-side and apex edges); for $M(G_3), M(G_4), M(G_5)$ all 34 single-element minors were certified individually. Since every proper minor of R is a minor of some single-element minor of R, and $\mathcal{M}_3$ is minor-closed [EGFS, Prop. 7.2], it follows that every proper minor of each $M(G_i)$ lies in $\mathcal{M}_3$; combined with $M(G_i) \notin \mathcal{M}_3$, each $M(G_i)$ is an excluded minor. The same fact gives pairwise independence: a non-member of $\mathcal{M}_3$ cannot be a proper minor of any $M(G_i)$ (all proper minors are members), so no one of the six non-members $M(K_{3,5}), M(G_1), \ldots, M(G_5)$ is a minor of another. For $M(G_1)$ the absence of a $K_{3,5}$ graph minor in G_1 was additionally confirmed by direct exhaustive check.

A strengthening: non-membership in $\widetilde{\mathcal{M}}_3$. Alongside $\mathcal{M}_\ell$, [EGFS, Rmk. 8.5] introduces the larger minor-closed class $\widetilde{\mathcal{M}}_\ell \supset \mathcal{M}_\ell$, obtained by asking only that the color profile be nonzero at some $s \in S$ rather than constant and nonzero, and observes that "it is easier to show that a matroid does not lie in $\mathcal{M}_\ell$ than it is to show that it does not lie in $\widetilde{\mathcal{M}}_\ell$." For graphic matroids of biconnected graphs the two tasks coincide, by a result of Engel and Schreieder: "For a biconnected graph G, any $\mathbb{F}_p$-solution of $\underline{R} = M(G)$ in $\mathrm{Alb}_p(\underline{R})$ has a constant color profile $(\lambda_s(b_s))_{s\in S} = (\lambda, \ldots, \lambda)$" [ES, Cor. 2.10], where biconnected means that $G - v$ is connected for every vertex v [ES, Prop. 2.9]. Each of $G_1, \ldots, G_5$ has vertex connectivity 3, in particular is biconnected. Suppose some $M(G_i)$ lay in $\widetilde{\mathcal{M}}_3$: it would admit an $\mathbb{F}_3$-solution with profile nonzero at some s, hence, by the corollary, with constant profile $(\lambda, \ldots, \lambda)$, $\lambda \neq 0$; since the solution condition and the color profile are linear in (b_s), scaling by λ^{-1} would yield a solution with profile $\equiv 1$ — precisely what the dual certificates above refute. Hence $M(G_i) \notin \widetilde{\mathcal{M}}_3$ for $i = 1, \ldots, 5$, and since every proper minor lies in $\mathcal{M}_3 \subset \widetilde{\mathcal{M}}_3$, each is an excluded minor for $\widetilde{\mathcal{M}}_3$ as well. This strengthening is the combination of [ES, Cor. 2.10] with the certificates of this note; it is not available from the certificates alone. For $M(K_{3,5})$ the same conclusion $M(K_{3,5}) \notin \widetilde{\mathcal{M}}_3$ follows from [ES, Thm. 1.7] alone (the case $p = 3$), whose statement gives the trivial-profile form directly.

Verification. The verdicts were produced by an independent reimplementation of the membership test built solely from the definitions of [EGFS] (Defs. 5.1, 5.3, 5.8, 8.3, 8.9), validated by reproducing every worked value stated in [EGFS] — the solution-space dimensions $15, 103, 35$ of $M(K_{3,3}), M(K_5), R_{10}$; the memberships and radical distances $d = 2$ of these three; $d = 1$ for cographic matroids; and $M(K_{3,5}) \notin \mathcal{M}_3$ as an excluded minor. Each of the five non-membership verdicts was reproduced by a second, independently written solver (a distinct coordinate model for the Albanese group, a distinct matrix-assembly order, and an independently implemented $\mathbb{F}_3$ rank computation cross-checked against a second linear-algebra engine), with exact-integer agreement on every recorded rank (the values tabulated above). Every verdict in this note is certified by an explicit witness — a dual certificate for each non-membership, a solution vector for each minor membership — verified by direct arithmetic over $\mathbb{F}_3$ without trusting any rank engine.

Code and data. The certificate bundle accompanying this note contains, for each of the five matroids, the dual certificate and the minor primal witnesses of this section (for $M(G_1)$ and $M(G_2)$, one primal witness per automorphism orbit, with the orbit accounting recorded in the bundle manifest; for $M(G_3), M(G_4), M(G_5)$, all 34 single-element minors individually), the published $M(K_{3,5})$ non-membership dual as a pipeline anchor, the graph data with integer realizations, and a single self-contained checker, `m3_check.py`, requiring only Python 3 with numpy. Each witness file carries the realization matrix A; the checker rebuilds every Albanese membership system from A alone and verifies each witness by integer matrix–vector products mod 3 — no rank engine, and no trust in the software that found the witnesses. The single command `python3 m3_check.py` re-verifies every claim of this section; the expected final output line is `RESULT: PASS` (measured: about one second and well under 1 GB of memory, with Python 3.13 and numpy 2.3). The checker also verifies, against graph6 strings embedded in the checker itself, that every realization is a reduced oriented incidence matrix of the named graph, and that the certified minors cover all single-element deletions and contractions. The $\ell = 5$ membership witnesses of §4 are archived alongside, with their own self-contained checker, in the bundle's `l5_membership/` directory. The bundle, together with this note, is archived at `https://github.com/thebookersmith/hodge-m3-excluded-minors`.

4 Membership at $\ell = 5$

The wider significance of the radical distance has recently been settled at the level of divisibility. Engel and Schreieder [ES] prove that for a closed subvariety Z of codimension k in a very general principally polarized abelian variety of dimension g, the coefficient m in $[Z] = m \cdot \theta_k$ (with θ_k the minimal class) is divisible by every prime $p \leq (k+1)/2$ ([ES, Thm. 1.1]); equivalently $d(M(K_n))$ is divisible by all primes $p \leq (n-1)/2$ and grows at least exponentially in n ([ES, Cor. 1.6]). The prime-p obstruction is realized by $M(K_{2p+1}) \notin \mathcal{M}_p$ and $M(K_{3,2p-1}) \notin \mathcal{M}_p$ ([ES, Thms. 1.5, 1.7]). For $p = 5$ the two smallest such witnesses are $M(K_{11})$ (rank 10, corank 45) and $M(K_{3,9})$ (rank 11, corank 16): a factor of 5 provably appears, but only via matroids of large corank. It is then natural to ask where, in terms of size, prime 5 first appears. We contribute the first data at the small-corank end.

At the time the $\ell = 5$ computation reported here was carried out, three excluded minors for $\mathcal{M}_3$ were known: $M(K_{3,5})$, $M(G_1)$, and $M(G_2)$ — all of corank 8.

Theorem 4.1. *Each of $M(K_{3,5})$, $M(G_1)$, and $M(G_2)$ lies in $\mathcal{M}_5$.*

Each verdict is certified by an explicit $\ell = 5$ solution — a primal witness $(b_s)_{s \in S}$ with $\sum_s u_s \, \partial b_s = 0$ for a basis of $\bar{U}$ and constant color profile $\lambda_s(b) = 1$ for all s — on the full, uncompressed membership system of [EGFS, Def. 8.3], with $5^8 = 390{,}625$ Albanese vertices and up to 7.03×10^6 colored edges:

matroid	(g, n)	$\lvert V\rvert = 5^8$	rows ($g\,5^8{+}n$)	cols ($n\,5^8$)	verdict
$M(K_{3,5})$	$(7, 15)$	390,625	2,734,390	5,859,375	$\in \mathcal{M}_5$
$M(G_1)$	$(8, 16)$	390,625	3,125,016	6,250,000	$\in \mathcal{M}_5$
$M(G_2)$	$(10, 18)$	390,625	3,906,268	7,031,250	$\in \mathcal{M}_5$

Every witness is checked by a self-contained verifier of the same design as the $\ell = 3$ checker (`l5_membership/l5_check.py` in the archived bundle): it rebuilds the Albanese membership system from the totally unimodular realization alone and requires only Python with numpy.

Consequences at $\ell = 5$. These appear to be the first $\ell = 5$ membership data on the corank-8 slice. Three of the six known excluded minors of $\mathcal{M}_3$ — including both members of the bipartite apex family — do *not* persist as obstructions at $\ell = 5$: each lies in $\mathcal{M}_5$, so $5 \nmid d(R)$ for each, and in particular neither $M(G_1)$ nor $M(G_2)$ yields $30 \mid d(8)$ or $30 \mid d(10)$. Prime-5 obstructions do exist in general — by [ES], $M(K_{11})$ and $M(K_{3,9})$ lie outside $\mathcal{M}_5$ — so this places only these three matroids on the member side. The $\mathcal{M}_5$-status of $M(G_3), M(G_4), M(G_5)$, the $\ell = 5$ case of Problem 8.8, the exact corank at which a factor of 5 first appears, and the $\ell = 7$ status of all six matroids remain open.

5 Remarks

Search procedure. All five witnesses arose from a systematic membership computation at $\ell = 3$ over a census of small non-planar graphic matroids of corank 8. The bipartite stratum was searched first (motivated by the bipartite shape of the known example $K_{3,5}$) and produced G_1; the same census's general stratum — with no structural restriction — subsequently produced G_3, G_4, and G_5; G_2 arose from the corank-8 census at 11 vertices. Within the strata searched to completion, corank ≤ 7 contained no non-members at $\ell = 3$ at all; the concentration of all known excluded minors at corank 8 reflects the census boundary, not an established structural fact. Likewise the search ordering reflects only heuristics; each matroid stands on its certificate alone. A separate account of the search methodology is given elsewhere.

Outlook. The known excluded minor $M(K_{3,5})$ sits at the head of the family $M(K_{3,n})$. Whether any of the five matroids of this note heads an infinite family of its own is an open follow-on question, not a claim made here; the structural diversity observed in §2 — two bipartite apex constructions, three sporadic non-bipartite graphs, and zero non-members among 214 constructed apex candidates — suggests only that the answer to Problem 8.8 will not be a single-family list, and the data do not yet support more than that.

The wider context has recently shifted. The two questions of [EGFS] on the size of the radical distance — whether it can be arbitrarily large (Problem 8.13, the "second-prime"/unboundedness question) and the growth of $g \mapsto d(g)$ (Problem 8.14) — have since been addressed by Engel and Schreieder [ES]: as recalled in §4, they prove $M(K_{2p+1}) \notin \mathcal{M}_p$ and $M(K_{3,2p-1}) \notin \mathcal{M}_p$ for every prime p, whence $d(M(K_n))$ is divisible by all primes $p \leq (n-1)/2$ and grows at least exponentially. This solves Problem 8.13 and partially answers Problem 8.14, and in particular shows that obstructions occur at every prime: primes beyond 3 do arise. What their argument does not do is exhibit excluded minors — it establishes non-membership without minimality — so Problem 8.8 remains open at every odd prime, and the witnesses of [ES] are large (for $p = 5$, $M(K_{11})$ of corank 45 and $M(K_{3,9})$ of corank 16). The natural refinements are therefore the identification of excluded minors at each prime and the location of the frontier at which each prime first appears; the $\ell = 5$ data of §4 is a first probe of the small-corank end of that frontier for $p = 5$.

Finally, a methodological remark. The certificate scheme used here decides a corank-8 instance at $\ell = 5$ in about eighty seconds of computation, by exploiting the automorphism group of the underlying graph to compress the membership system before solving. For $M(K_{3,9})$ at $\ell = 5$ (corank 16), where the 5^{16}-vertex system cannot be materialized at all, a different route succeeds: an explicit Farkas dual supported on a socle window of $\mathbb{F}_5[t_1, \ldots, t_{16}]/(t_i^5)$, giving a certificate-level verification of the non-membership underlying [ES, Thm. 1.7] at $p = 5$. The window certificate and its self-contained checker are included in the archived bundle (`window_certificates/`); the method itself will be described elsewhere.

References

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